

NEW BRUNSWICK COMMUNITY COLLEGE

CIVIL / BUILDING ENGINEERING TECHNOLOGY

CALCS. BY: Maksim Khavrai
DATE: Sept 19 2026

COURSE: PROJ1110
PROBLEM: Ex5

PAGE: 1

Petitcodiac Plaza PROJ1110.001
Gross Square Feet (GSF) =
= 100,000 SF

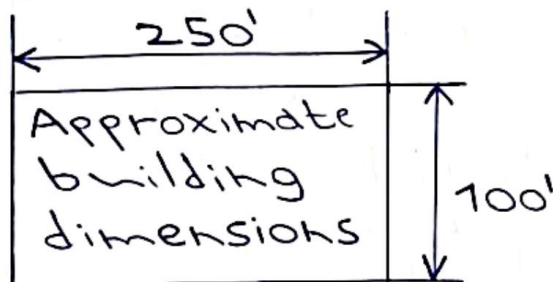
4 Levels (3 Levels above grade
+ U/G Parking)

Area per Floor (Footprint) =
= $\frac{\text{Gross Area}}{\# \text{ Levels}} = \frac{100,000 \text{ SF}}{4} =$
= 25,000 SF/Level

Structural Bays Size

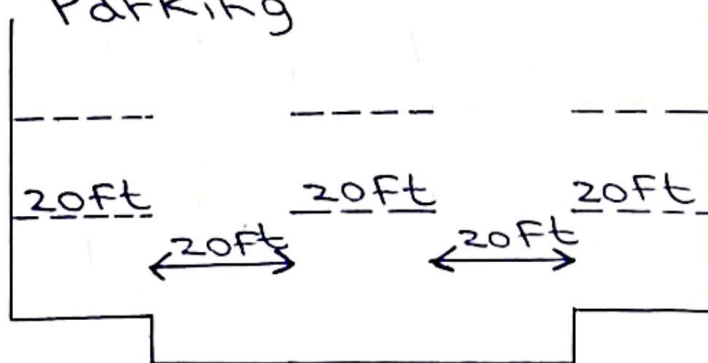
Max 8m span

8m X 3.28 Ft/m = 26.2 Ft
(max)



$25,000 \text{ SF} / 100' =$
 $= 250'$

Parking



Parking
aisle 18'-22'
Pricing
database
20' or 25'
Use 20ft
bay

NEW BRUNSWICK COMMUNITY COLLEGE
CIVIL / BUILDING ENGINEERING TECHNOLOGY

CALCS. BY: Maksim Khavrai
DATE: Sept 19 2026

COURSE: PROJ1110
PROBLEM: Ex5

PAGE: 2

Parking requires 5 Bays wide

$$5 \text{ bays} \times 20 \text{ ft/bay} = 100 \text{ ft wide}$$

Building length

$$\frac{25000 \text{ SF}}{100 \text{ ft}} = 250 \text{ ft} / 20 \text{ ft/bay} =$$
$$= 12.5 \text{ bays} \downarrow = 12 \text{ bays}$$

$$12 \text{ bays} \times 20 \text{ ft/bay} = 240 \text{ ft}$$

Length

$$\text{GSF} = 100 \text{ ft} \times 240 \text{ ft} \times 4 \text{ levels} =$$
$$= 96,000 \text{ SF}$$

Area per Floor (Footprint) =

$$= \frac{\text{Gross Area}}{\# \text{ Levels}} = \frac{96,000 \text{ SF}}{4} =$$

$$= 24,000 \text{ SF/level}$$

Useable space ratio =

$$= \frac{1.3 + 1.3 + 1.2 + 1.2}{4} = 1.25$$

$$\text{Useable space} = \frac{\text{Gross Area}}{\text{US ratio}} =$$

$$= \frac{24,000 \text{ SF/level}}{1.25} = 19,200 \text{ USF/level}$$

NEW BRUNSWICK COMMUNITY COLLEGE
CIVIL / BUILDING ENGINEERING TECHNOLOGY

CALCS. BY: Maksim Khavrai
 DATE: Sept 19 2026

COURSE: PROJ1110
 PROBLEM: Ex5

PAGE: 3

$$\text{Core Area} = 24,000 \text{ SF/Level} - 19,200 \text{ SF/Level} = 4,800 \text{ SF/Level}$$

$$\text{Length} = \frac{\text{Core Area}}{\text{Width}} = \frac{4800 \text{ SF/Level}}{20 \text{ ft}} = 240 \text{ ft} = \text{total length}$$

240 ft
not good

$$\frac{4800 \text{ SF/Level}}{30 \text{ ft}} = 160 \text{ ft}$$

$$160 \text{ ft} / 20 \text{ ft/bay} = 8 \text{ bays length}$$

$$30 \text{ ft} / 20 \text{ ft/bay} = 1.5 \text{ bays wide}$$

Tenant space design		☒ Ceiling height
Cafeteria - 7,282 (7,200)		10'
Childcare Center - 11,612 (11,600)		9'
Conference Rooms - 10,661 (10,600)		12'
Health Clinics - 1,840 (1,800)		9'
Offices - "Fully" Open - 20,572 (20,500)		9'
Physical Fitness Center - 7,076 (7,000)		12'
Retail Outlets - 2,714 (2,700)		12'

NEW BRUNSWICK COMMUNITY COLLEGE

CIVIL / BUILDING ENGINEERING TECHNOLOGY

CALCS. BY: Maksim Khavrai
DATE: Sept 19 2026

COURSE: PROJ1110
PROBLEM: EX5

PAGE: 7

Table 1 - Building Program Summary
(Standard of Acceptance)

Floor	Use 1	SF	Use 2	SF	Use 3	SF	Tenant Space (SF)	Common Core (SF)	Total Gross (SF)
Level 3	Offices	17,400	Health Clinics	1,800			19,200	4,800	24,000
Level 2	Coffee- house Rooms	11,500	Physi- cal Fitness Center	7,700			19,200	4,800	24,000
Level 1	Retail out- lets	2,700	Ca- re- center- rid	6,500	Child- care Center	10,000	19,200	4,800	24,000
U/G	Par- king	19,200					19,200	4,800	24,000
						Totals	76,800	19,200	96,000

NEW BRUNSWICK COMMUNITY COLLEGE
CIVIL / BUILDING ENGINEERING TECHNOLOGY

CALCS. BY: Maksim Khayrai
 DATE: Sept 19 2026

COURSE: PROJ1110
 PROBLEM: Ex5

PAGE: 5

**Table 2 - Building Elevation
 Preliminary Calculations
 (Standard of Acceptance)**

Description	Δ Elevation	Actual Elevation
Roof		43.2'
Structure	+1.7'	+43.2'
MEP	+1.7'	+41.5'
Ceiling Aff Max	+ 9'	+39.8'
Level 3		30.8'
Structure	+1.7'	+30.8'
MEP	+1.7'	+29.1'
Ceiling Aff Max	+ 12'	+27.4'
Level 2		15.4'
Structure	+1.7'	+15.4'
MEP	+1.7'	+13.7'
Ceiling Aff Max	+ 12'	+12'
Level 1		0.0'
Structure	- 1'	- 1'
MEP	-1.75'	-2.75'
Clearance	- 7'	-9.75'
Level 0 (U/G)		-9.75'

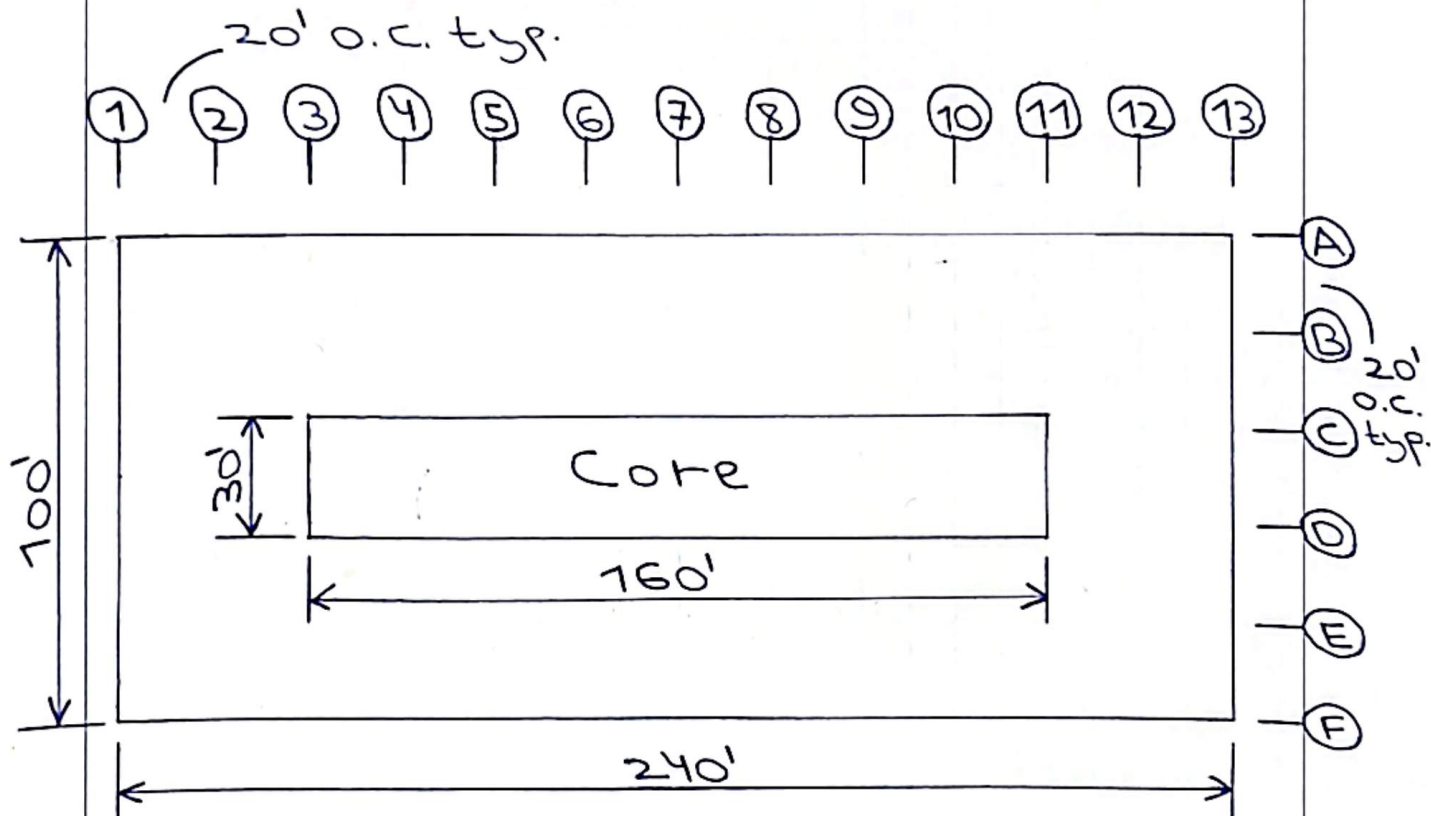
NEW BRUNSWICK COMMUNITY COLLEGE
CIVIL / BUILDING ENGINEERING TECHNOLOGY

CALCS. BY: Maksim Khavrai
DATE: Sept 19 2026

COURSE: PROJ1110
PROBLEM: Ex5

PAGE: 6

Proposed Floor + Grid layout



$$\text{Gross Area Building} = 100' \times 240' = 24,000 \text{ SF}$$

$$\text{Area Core} = 30' \times 160' = 4,800 \text{ SF}$$